Short-term gains vs. long-term Success: Reward strategy design for reinforcement learning in football

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***Abstract*— Reinforcement learning in complex games like soccer relies heavily on how you define your reward function and environment. In this work, we developed a custom 3v3 soccer environment and implemented two RL-based teams with distinct learning trends: one with a fast convergence but limited long-term adaptation, and another with a slower yet more robust learning trajectory. Simulation shows that despite performing better at the start, the short-term agents fall short of the performance of the long-term agents in the long run, and after passing 50% of the episodes, the win rate of long-term agents rises from 30% in the beginning to 50%.**

Keywords*— *Reinforcement Learning, Multi-agent systems, Soccer Simulation.

I. Introduction

The usage of Artificial Intelligence as an agent in real-life simulations is increasing day by day. From self-driving cars [1] to robotics [2], to stochastic games like Go [3], Chess [4], and recently Soccer [5].

The universally popular sport called soccer, as a stochastic multi-agent game, is observable as a conditional problem, where the related conditions are the position of the ball, the rivals' goal and the other players on the field [6]. Hence, designing simulation environments and models has seen increasing attention in analytics and simulation as in computer games [7], and assistance to coaches of the sport [8].

Reinforcement Learning (RL) has been popularly used in training agents which are active in unfamiliar environments [9], since RL is a great unsupervised learning method for complicated conditional programs with unknown dynamics based on rewards that are set in the policies of the problem.

In reinforcement learning, the definition of the reward function plays a crucial role in shaping the agent's behavior and directly affects the overall performance of the system. Poorly designed rewards can lead to undesired behaviors or reward hacking, whereas a carefully structured reward can significantly accelerate learning and improve generalization. Therefore, reward design is considered one of the most critical challenges in applying RL to complex domains such as multi-agent sports simulations [10][11].

In this research, we implemented and compared two distinct reward schemes for training RL agents in a simulated soccer environment. The first scheme was based solely on primary outcomes, where agents were rewarded for scoring a goal and penalized for conceding one. The second scheme included secondary performance indicators, such as successful passes and shots on target, alongside the primary goal-related rewards. This comparison allowed us to evaluate whether shaping the reward with additional intermediate events could lead to faster learning dynamics and more tactical behavior compared to relying only on sparse, outcome-based rewards.

II. Methodology

A. Environment

Currently, there are few open-source football simulation environments that allow integration of custom controllers, and those that exist are not well-suited to our objectives. Our focus is on developing and testing tactical strategies rather than engaging with graphically intensive football environments. For this reason, we designed and implemented a custom environment using the *PyGame Python* library. During reinforcement learning training, the environment can also be executed in headless mode (without graphics) to improve efficiency and the simulation frequency is reduced from 0.3 to 0.1.

The developed environment is simple and low in graphical complexity, fitted to meet our research goals. It supports essential functions such as shooting, passing to another player (rather than to a specific position), and movement. In addition, we implemented logical constraints to prevent unrealistic combinations of actions, such as performing a pass and a shot simultaneously or executing opposed moves like moving upward and downward at the same time.

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To enhance realism, the environment incorporates features such as the gradual reduction of ball speed during passes and shots.

We have 3vs3 games, and for the sake of simplicity, both goalkeepers are automated and not controlled by either controller; they move with respect to the position of the ball. The environment interacts with external controllers by writing game data into a database and retrieving control actions from it. In this study, two types of controllers are used: reinforcement learning based on Proximal Policy Optimization (PPO) and imitation learning.

Figure 1 shows the graphical interface of the football simulator. Initially, the environment was visualized in this graphical form, but in later stages it was run in a headless mode to enable faster training and data collection without rendering overhead.

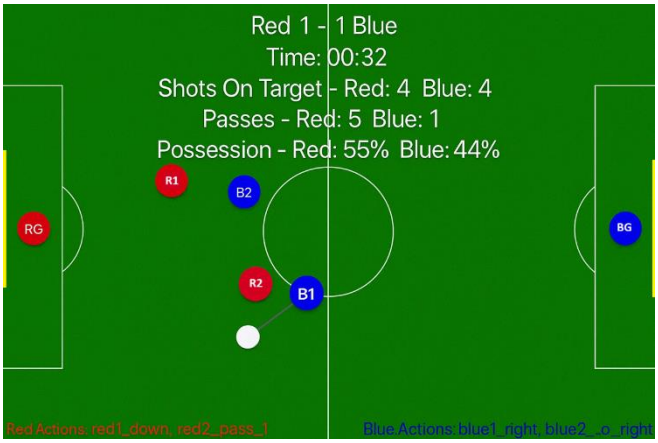

Figure 1. *The environment setup of the football simulator.*

Figure 2 presents the overall architecture of the game framework. The environment communicates with both the red and blue control databases, which in turn interact with their respective Proximal Policy Optimization (PPO) strategies. The short-term and long-term strategies update the controllers, and all interactions are logged in the game database for subsequent analysis. This structure highlights how the training loop is organized and how different components exchange information throughout the learning process.

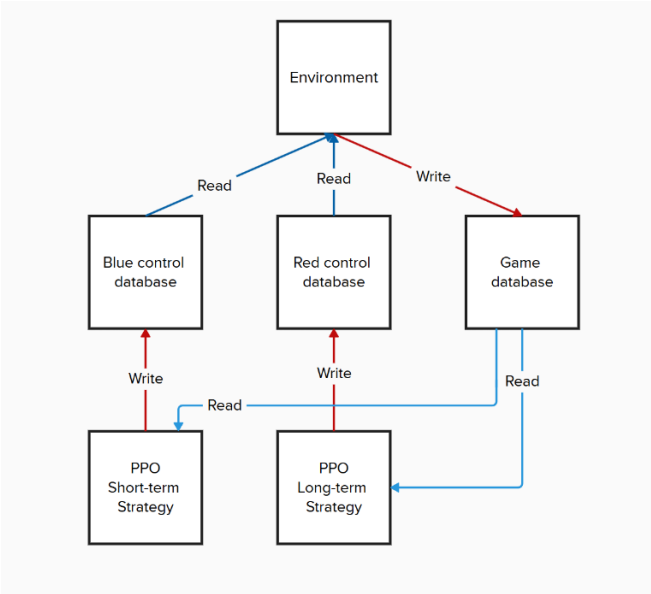

Figure 2. *The architecture of the football simulation framework illustrating the interaction between the environment, control databases, and PPO-based strategies (short-term and long-term) through read/write operations.*

B. RL Agents

For the reinforcement learning component, we employed **Proximal Policy Optimization (PPO)** as the primary algorithm. PPO was selected due to its greater stability and robustness compared to alternative algorithms; a characteristic also demonstrated in the Google Research Football framework. This stability makes PPO particularly effective for environments where training dynamics are complex and may otherwise lead to divergence or instability.

To shorten training time and improve overall efficiency, significant effort was made to minimize the dimensionality of the state space. By carefully reducing the state representation to only the most relevant features, we were able to accelerate convergence while preserving the essential information required for tactical decision-making.

As previously described, each team consists of three players, with the goalkeeper fully automated and not controlled by any learning algorithm. The remaining two players are controlled through a mixed strategy combining **rule-based logic** and **reinforcement learning (RL)** via PPO. At any given timestep, only one of the two field players is controlled by the RL agent, while the other follows rule-based behavior.

The decision of which player is assigned to the RL controller is determined by the following rules:

- When the team has the ball, the player currently holding the ball is controlled by the RL agent.

- When the team does not have the ball, the player positioned farther from the ball is controlled by the RL agent. This design enables the agent to learn off-

the-ball movements, such as improving defensive positioning or optimizing space occupation.

The rule-based controller implements simple, deterministic behaviors.

- If the ball is not in possession, the controlled player attempts to move toward the ball.
- If the team is in possession, the rule-based player adapts its movement relative to the RL-controlled teammate, such as moving forward to support an attack or retreating to maintain defensive structure.

- ***Reward Function***

The reward function is applied only to the RL-controlled agent at each timestep. It is designed to balance individual and team objectives. We have trained 2 RL controllers that differ from each other just in the Reward functions, which consist of the following components:

Long-term strategy:

- **10 Positive reward** for scoring a goal.
- **5 Negative reward** for conceding a goal.
- **0.5 Positive reward** for successful passes.
- **0.5 Positive reward** for taking a shot.

Short-term strategy:

- **10 Positive reward** for scoring a goal.
- **5 Negative reward** for conceding a goal.

- *Proximal Policy Optimization Configuration*

The configuration parameters used for PPO in our experiments are summarized in **Table 1**.

Table 1. The configuration parameters.

Parameter	Value
Action Space	11
Actor LR	3×10^{-4}
Critic LR	1×10^{-3}
Discount Factor (γ)	0.99
GAE λ	0.95
Clip ε	0.20
Entropy Coef.	0.01
Value Coef.	0.50
Batch Size	256
Rollout Horizon	512
Training Epochs	4

- ***State and Action Spaces***

The definition of the state space and action space used in the environment is provided in **Tables 2 and 3.**

Table 2.Space state parameters.

	Feature	Normalization
0–1	Player1 (x, y)	÷ (800, 500)
2–3	Player 2 (x, y)	÷ (800, 500)
4–5	Opponent player 1 (x, y)	÷ (800, 500)
6–7	Opponent player 2 (x, y)	÷ (800, 500)
8–9	Ball (x, y)	÷ (800, 500)
10	goals	÷ 20
11	Opponent goals	÷ 20

Table 3.Action space parameters.

ID	Action
0	No-op
1	Up
2	Down
3	Left
4	Right
5	Up Right
6	Down Right
7	Up Left
8	Down Left
9	Shoot
10	Pass

C. Criteria and specification of the experiment

In the initial setup, the frequency of reading and writing actions and game data was set to **0.3.** To improve training efficiency, the environment was later executed in **headless mode**, and the frequency was reduced to **0.1**.

Both the reinforcement learning controllers (based on PPO) were executed concurrently with the environment to enable real-time interaction. Each match lasted **five minutes**, after which the simulation was terminated and restarted in a loop to allow continuous training and evaluation.

For this research, two reinforcement learning (RL) agents against each other (Long-term strategy and Short-term strategy) for 2,900 episodes have been trained. During each episode, detailed logs were recorded into an Excel file. These logs included:

- Neural network parameters and coefficients used during training,
- The number of shots, passes, and goals for each team,
- Additional performance indicators relevant to tactical evaluation.

This configuration ensured systematic monitoring of both the learning process and the overall performance of the agents across multiple episodes.

III. Results

As said before, we have trained both Reinforcement Learning controllers with different reward strategies against each other for 2900 games to contrast the differences between the 2 strategies. In Figure 3, we analyze the win rate of the two controllers, which provides the most reliable basis for comparison. Following that, Figure 4 presents the number of goals achieved under each strategy, which further supports and validates the observations from Figure 3. The summarized results are as follows:

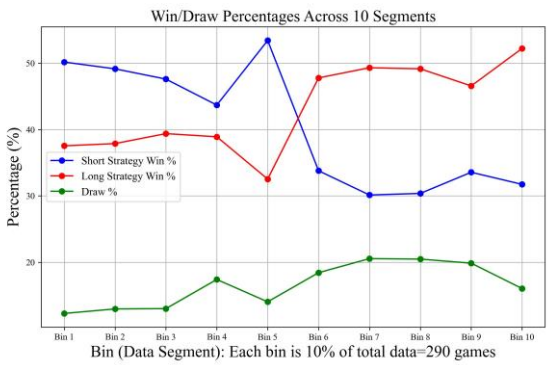

Figure 3. *Percentage of wins and draws for Reinforcement Learning controller with long-term strategy and short-term strategy.*

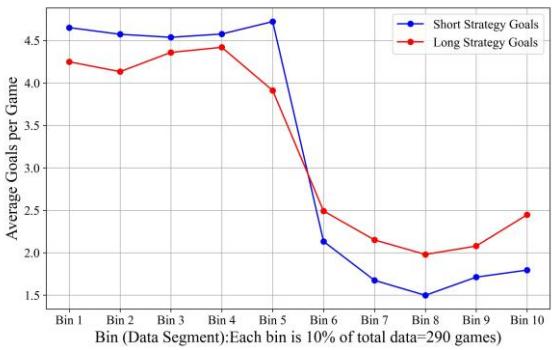

Figure 4. *Average of total goals for Reinforcement Learning controller with long-term strategy and short-term strategy.*

Due to differences in frameworks and methodologies, the results are not directly comparable with those reported in other studies. Moreover, most related works have used the GRF (Google Research Football Team) environment, and none of them have trained two reinforcement learning agents against each other.

However, to examine the algorithm and the reward design, the results can be qualitatively compared with part of Google's work. In their study, the algorithm that assigned rewards not only to goals scored and conceded but also to passes and shots demonstrated better progress. It should be noted that Google's RL-based controller was compared against a rule-based controller, and the environments were substantially different. Therefore, the results are not quantitatively comparable; nevertheless, from a qualitative perspective, the pattern observed in Google's findings was reproduced here as well.

IV. Discussion

The analysis of the charts highlights important differences in performance between the Blue Team, which focused on a short-term strategy based only on goals scored and conceded, and the Red Team, which followed a longer-term approach that also rewarded passes, shots, and overall gameplay quality.

At the start, the Blue Team appeared stronger, achieving nearly 50% wins in the first bin. This suggests that concentrating on immediate goal outcomes offered an early advantage, likely because of its direct alignment with the match objective. In contrast, the Red Team started more slowly, with win rates below 40% in the first bin, as their strategy emphasized building up play through passes and shots before converting these into goals.

A clear turning point came around Bin 5, when the Red Team's win percentage began to surpass the Blue Team's, eventually reaching about 50% in Bin 10. This shift indicates that the Red Team's long-term strategy paid off, enabling them to gain an edge through sustained performance rather than quick results.

The percentage of draws stayed relatively low and stable (around 10–20%), showing that most matches had clear winners. Across nearly 2,900 five-minute games, representing roughly 87 million logged actions, the dataset provides a strong basis for these observations.

The average goals per game further support these findings. The Blue Team consistently scored between 4.0 and 4.5 goals per game, reflecting their focus on direct outcomes but limited progression. The Red Team's scoring fluctuated, dipping to around 1.5 goals in Bin 6 before recovering to roughly 2.5–3.0 goals in later bins. This pattern suggests a period of adjustment, during which the Red Team refined its strategy, ultimately leading to greater scoring efficiency. The drop in the number of goals for each team shows that both teams have learned how to defend better.

Overall, the results show that while short-term strategies can deliver quick wins, a broader, long-term approach that values overall game performance can create more sustainable success. Future studies could explore how different action frequencies and reward weightings influence the balance between short-term gains and long-term advantages.

V. FUTURE WORKS

In our future works, we plan to expand our experiments to larger team sizes (e.g., 5v5 and 11v11) and use other types of learning like imitation learning, as it would replicate the real players' behavior more directly. Additionally, we aim to refine the reward structure to capture more complex tactical behaviors and to introduce advanced performance metrics beyond match outcomes. These directions could help develop more realistic agents, ultimately contributing to improved strategies in soccer and related domains.

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